Differential entropy

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This page extends the surprisal and entropy of discrete random variables to continuous random variables.

Surprisal and entropy of a continuous random variable

Definition 0.1 (Surprisal of a value of a continuous random variable)**.** The **surprisal** of a possible value x of a continuous random variable X with density f and $f(x) > 0$ is

$$s(x) \stackrel{\text{def}}{=} -\log f(x).$$

This extends the surprisal of a value of a discrete random variable, with the density $f(x)$ in place of the probability $\mathbb{P}(X = x)$.

Definition 0.2 (Differential entropy)**.** The **differential entropy** of a continuous random variable X with density f is the expectation of its surprisal, when that expectation exists (Cover and Thomas 2006, sec. 8.1):

$$h(X) \stackrel{\text{def}}{=} \mathbb{E}[s(X)] = -\int_{\{x:f(x)>0\}} f(x) \log f(x)\, dx.$$

Example 0.1 (Differential entropy of a uniform random variable)**.** Let X be uniformly distributed on the interval from 0 to $a > 0$, so that its density is $f(x) = 1/a$ for $0 < x < a$ and 0 elsewhere. Every value in the interval has surprisal $s(x) = -\log(1/a) = \log a$, so the differential entropy is $\log a$ (Cover and Thomas 2006, Example 8.1.1):

$$\begin{aligned} h(X) &= -\int_0^a \frac{1}{a} \log \frac{1}{a}\, dx \\ &= \log a. \end{aligned}$$

With $a = 1/2$ and base-2 logarithms, $h(X) = -1$ bit.

Remark 0.1 (Differential entropy differs from entropy). A density value is not a probability, so the surprisal of a continuous value and the differential entropy behave differently from their discrete counterparts:

- **Sign.** A density can exceed 1, so the surprisal $s(x)$ and the differential entropy can be negative, as in Example 0.1 with $a < 1$ (Cover and Thomas 2006, sec. 8.1). The entropy of a discrete random variable is never negative.
- **Units.** Rescaling changes the differential entropy: $h(aX) = h(X) + \log|a|$ for a constant $a \neq 0$ (Cover and Thomas 2006, Theorem 8.6.4). Measuring the same quantity in metres rather than centimetres therefore changes h. Adding a constant does not: $h(X + c) = h(X)$ (Cover and Thomas 2006, Theorem 8.6.3).
- **Not a limit of entropy.** Round X to bins of width Δ, which gives a discrete random variable X^Δ. Then $H(X^\Delta) + \log\Delta \to h(X)$ as $\Delta \to 0$ (Cover and Thomas 2006, Theorem 8.3.1). The entropy $H(X^\Delta)$ itself grows without bound as $\Delta \to 0$, so $h(X)$ is not its limit.

References

Cover, Thomas M., and Joy A. Thomas. 2006. *Elements of Information Theory*. 2nd ed. Wiley-Interscience. https://doi.org/10.1002/047174882X.